

Predicting Hit Songs Using Audio Features and Social Media Metrics

Jonathan Leewin

Computer Science Department

School of Computing and Creative Arts

Justin Hadinata

Computer Science Department

School of Computing and Creative Arts

Ari Jaya Teguh

Computer Science Department

School of Computing and Creative Arts

Nunung Nurul Qomariyah

Computer Science Department

School of Computing and Creative Arts

Raymond Bahana

Computer Science Department

School of Computing and Creative Arts

Abstract—The changing nature of music has brought better ways to forecast future song trends. Previous techniques involved simple, auditive characteristics like tempo and key, or, more recently, statistical methods to anticipate which songs could become popular ones. Lately, social networks have predominated in the music industry, deepening the parameters for estimating conquering popularity. This paper examines the relationship between audio features in a song and an artist's social media influence to improve the predictive model. To that end, for the most-streamed songs of each year 2023 and 2024 datasets with audio features and social metrics were gathered. The tested Machine learning models include Random Forest, Gradient Boosting, Support Vector Machines, and Fully Connected Neural Networks. Research findings show that using features of social media integrated with audio data enhances the effectiveness of the model to gain better accuracy (81.26%) and lower error margins. These findings provide some insights that suggest that extending variables from outside the predicted models, especially social media relevance, should be integrated for forecasting the popularity of music.

Index Terms—Music Popularity Prediction, Audio Features, Social Media Metrics, Machine Learning, Spotify Dataset, Random Forest, Neural Networks, Gradient Boosting, Predictive Analytics, Music Industry

I. INTRODUCTION

Music, a universally known concept that dates back as far as we can imagine, has evolved rapidly throughout the centuries. Previously, potential hit songs could be predicted using traditional data such as acoustic features including tempo, key, and harmonies. Together with this, statistical models can show how the track might perform in popularity [1]. However, more recently, there have been changes as to who and what predicts music. Social networks have undoubtedly had a significant impact on our society, especially in the music industry. The

interaction with the artist's audience and the presence of social media platforms such as social media provided a space to select the probability of the viral potential of the song. This new variable has allowed the possibility of predicting potential popular tracks with a higher degree of precision.

II. LITERATURE REVIEW

Music is one of the most sought-after entertainment. Musicians compete to create their masterpieces and entertain their listeners. Studies even analyze what components or essential things are needed to determine a certain song's success. There are several components that have an effect on making the song more popular. One of which is using audio features from Spotify. Spotify judges a song based on its tempo, energy, valence, etc.

A study by Saragih discusses how the Spotify features affect the popularity of the songs. Using multiple classifiers, he came to the conclusion that audio features have an impact on the popularity of songs [2]. Another study has a topic similar to that in the first paper. However, the study by Yee and Rahim addresses that in addition to Spotify audio features, YouTube metrics have an effect on the popularity of songs [3]. This shows that features and popularity on streaming platforms such as YouTube have a significant impact on the success of the songs.

Besides using audio features, there is a study that approaches this problem by using Music Charts. The study discusses how being popular on a streaming platform such as Spotify has a huge impact on the popularity of the song [4]. By using machine learning, the study succeeds in predicting the popularity of a certain song in advance [4]. Another study talks about how music sales and artist popularity can represent how popular the song is currently. Having a large amount

of followers on social media highly affects how successful a song will be [5]. Lastly, a more complex study addresses that it is feasible to predict a song's popularity using complexity, MPEG-7, and Mel-frequency cepstral coefficient (MFCC) features and MFCC features. By comparing the popularity of the songs before, during, and after their debut, the studies can predict the popularity of the song [6].

Collectively, all of these studies illustrate how the success of a song can be impacted by a great number of things. It is affected by audio features, social media metrics, and how well the sales of that song are. Each of the studies has its uniqueness and gives a brand new explanation of the problem. However, every study has the limitation of not considering other variables that affect the popularity of the songs. To address this limitation, our study offers to predict the popularity of the song using its audio features and the artist's social media metrics. This study will help bridge the gap, which is the correlation between the audio characteristics of the song and the popularity of the main artist who creates or sings the song.

III. METHODOLOGY

A. Dataset Overview and Description

This research consist of two datasets. Both data sets incorporate detailed features of the most streamed Spotify songs. The first dataset is for the most streamed songs in 2023, while the second one is in 2024. Both of the data sets are obtained through a popular data science competition platform called Kaggle. At first, we attempted to obtain our own data through scraping data from social media and Spotify. However, after weeks of attempts to scrape from social media, the challenges and obstacles are proven to be difficult for us to handle so we resolve utilize Kaggle instead. After a few consideration, we conclude that make use of Kaggle is adequate as long as the datasets are made in high quality, fit within our research objective, and captures the relevant timeframe.

The time frame capture is not only the difference between the two datasets, as the features within these datasets also differ. The 2023 data sets primary in-capture the audio features of the songs while also still consist of data that in-capture the song's popularity within different platforms. The second data set primary incorporates information regarding the song's popularity overall across different platforms and does not touch the audio aspects of the song. This information is crucial to better understand the correlation between the popularity of the songs and the audio features. We use only numerical values for our data.

B. Dataset Preprocessing

The first data, 2023 most streamed songs, includes details of the songs, popularity, and audio features. The information of the songs consists of 24 columns, most of the columns are divided into two categories which are the audio features of the songs and the popularity aspect. The data set consists of more than 954 rows. The other dataset, 2024 most streamed songs,

includes details of the songs, popularity, and social media metrics. This data set consists of 29 columns and primarily discusses the prevalence of songs across different platforms such as Youtube, Tiktok, Spotify, etc. This data set consist over 4601 rows. Both data sets are combined into one CSV file that contains both the audio features and the social media metrics that are needed for this research paper.

The final data consists of 35 columns that incorporate the necessary information for our machine learning models. The features are track name, artist(s) name, artist count, released year, released month, released day, in spotify playlists, in spotify charts, streams, in apple playlists, in apple charts, bpm, mode, danceability, valence, energy, acousticness, instrumentalness, liveness, speechiness, Track, Album Name, Artist, Release Date, ISRC, All Time Rank, Track Score, Spotify Streams, Spotify Playlist Count, Spotify Playlist Reach, Spotify Popularity, YouTube Views, YouTube Likes, TikTok Posts, TikTok Likes, and TikTok Views.

To ensure that the data are sufficient for our machine learning algorithms, we implement some necessary steps to process and clean our data. The first thing we do is replace null values with values we deemed fit through our domain knowledge or completely remove the problematic attributes. After ensuring that our data set does not have missing values and can be properly used, we then started to separate the data to training and test sets. For some machine learning models that are particularly sensitive to the scale of the data such as CNN, we applied normalization for the dataset.

IV. RESULTS AND DISCUSSION

A. Models Evaluated

This research focuses on machine learning models which have been applied to calculate the optimal prediction for each corresponding data, as well as their performance in handling complex data.

Random Forest is a mention of this models, which is a machine learning method that ensemble a collection of decision trees $T_1, T_2, \dots, T_n$ during training. The data is partitioned randomly into n trees to which each tree is trained. In the case of regression or majority voting for classification, the final prediction is obtained by averaging the predictions. The model can be represented as:

$$\hat{y} = \frac{1}{n} \sum_{i=1}^n T_i(x)$$

where $T_i(x)$ is the prediction of the i -th tree for input x [7].

Gradient Boosting is a machine learning that builds up the trees, with each tree sequentially correcting the errors (residuals) of the previous one, through a boosting technique. We denote by $r_t = y - f_{t-1}(x)$ with the residuals at iteration t , where $f_{t-1}(x)$ is the prediction of the previous ensemble. A new tree $T_t(x)$ is fitted at each iteration to these residuals. The final prediction is the sum of all trees' outputs:

$$f(x) = \sum_{t=1}^T \eta T_t(x)$$

where η is the learning rate, $T_t(x)$ is the prediction from the t -th tree, and T is the total number of trees [8] [9] [10].

Support Vector Machine is a classifier which finds the separating hyper plane that creates the maximum margin (distance) between the hyper plane and the nearest data points (support vectors). The optimization problem is formulated as:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to the constraints:

$$y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i \quad \text{and} \quad \xi_i \geq 0$$

where $\mathbf{w}$ is the weight vector, b is the bias, ξ_i is the slack variable for the i -th data point, and y_i is the label of the i -th sample. [11]

A Fully Connected Neural Network has an input layer, and an output layer along with some or no number of hidden layers. Minimizing the loss function L , cross entropy loss frequently or some other defined function, the network learns. The forward propagation in an FCNN can be expressed as:

$$\mathbf{h} = f(W\mathbf{x} + b)$$

where $\mathbf{x}$ is the input vector, W is the weight matrix, b is the bias vector, and f is the activation function (e.g., ReLU, sigmoid). The model's final prediction is obtained after passing the input through all layers and computing:

$$\hat{y} = g(W_{\text{out}}\mathbf{h} + b_{\text{out}})$$

where g is the output activation function, and W_{out} and b_{out} are the final layer's weights and bias. [12] [13]

B. Feature Sets

Two feature sets were used to train the models:

- **Audio Features Only:** These characteristics of the audio are derived from extracted tracks including tempo, key, and other spectral features.
- **Audio Features + Social Media Metrics:** Alongside its audio features, various social media metrics including number of shares, likes and comments were included to also consider externally driven popularity.

C. Performance Metrics

The performance of the models was evaluated using the following metrics:

- **Accuracy:** The ratio of correct predictions to total predictions [14]:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}} \quad (1)$$

D. Comparative Analysis of Model Performance

The models' performance was evaluated under two feature sets: Both audio features only, and audio features combined with social media metrics. Table I summarizes the results.

TABLE I
MODEL PERFORMANCE COMPARISON: AUDIO FEATURES ONLY VS.
AUDIO + SOCIAL MEDIA METRICS

Model	Accuracy
Random Forest (Audio Features)	58.65%
Gradient Boosting (Audio Features)	57.89%
Support Vector Machine (Audio Features)	61.65%
Fully Connected Neural Network (Audio Features)	61.94%
Random Forest (Audio + Social Media)	81.20%
Gradient Boosting (Audio + Social Media)	80.45%
Support Vector Machine (Audio + Social Media)	77.44%
Fully Connected Neural Network (Audio + Social Media)	81.26%

1) Performance Gains from Social Media Metrics:

The effect of incorporating social media metrics is quantified as follows:

$$\text{Accuracy} = 81.26\% - 61.94\% = 19.32\%$$

These improvements emphasize the value of integrating social media metrics, as they account for external influences on a song's popularity that audio features alone cannot capture.

E. Advantages of Integrated Social Media Metrics

The results of this study show that using social media metrics with audio features yields substantial advantages in terms of better prediction of songs' popularity. Of course, increase in prediction accuracy is one of the most significant benefits of applying the framework. We observe that the integration of social media measurements yields significant improvements in accuracy for all machine learning strategies. For example, the Fully Connected Neural Network has the Δ Accuracy improvement of 19.32%; therefore it is clearly evidenced that using of the external factors produces positive results.

Another major strength is also the possibility to evaluate external factors affecting a song's popularity. To some extent, the number of likes, shares, and comments, respond to cultural and social factors that cannot be captured by the features of audio. It is made possible for models to consider other factors which led to a song becoming popular other than the mere calibration of a song.

It is also important to note that the use of audio features and social media metrics has high real-world practical relevance. Since the popularity of a song may rely on the ability to be human- interactively present on the World Wide Web, this integration of the models brings the models closer to reality. Therefore, these models prove superior to predicting top songs in real world.

Moreover, this approach has utility to be used in other fields and not just limited to accounting. More precisely, due to the utilization of distinct feature repertoires, the models can be employed not just for estimating popularity, but also for providing recommendations or designing marketing campaigns. To other stakeholders like music producers, marketers, these

insights make it possible to make decisions and promote a song and get maximum results.

V. DISCUSSION

A. Analysis of Results

These results highlighted in Table I further strengthen the potential of incorporating social media metrics into machine learning model for predicting the popularity of songs. Integration of both audio features and social media metrics allowed for considerably higher accuracy compared with the indicators of separate models; the highest accuracy was noted in the Fully Connected Neural Network (81.26%). This improvement shows that although audio features themselves are useful for modelling the dynamics of music popularity, it is not sufficient to use them alone.

The better results indicate the possibility of FCNN to capture interactions between multiple features where a lot of feature space is nonlinear, time series of social activity, and acoustic characteristics of songs. Random Forest and Gradient Boosting models, which performed well when trained with audio features, did not harness the integrated dataset entirely well – their performance was slightly inferior to the performance of the FCNN.

B. Significance of Social Media Metrics

As has been demonstrated in this analysis, it is critical to include both social media features and audio features in machine learning models. Integration of social media metrics has remained statistically significant in each of the analyzed models, and consistently so. Therefore, from the results obtained using the paired t-test, there was a significant improvement on accuracy. The calculations and results are presented below:

1) *Paired t-Test Formula*: The analysis of variance was also performed to find out if the inclusion of social media metrics has resulted in significant differences. The formula for the paired t-test is as follows:

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

Where:

- $\bar{d}$: Mean of the differences between the two conditions (Audio-only vs. Audio + Social Media)
- s_d : Standard deviation of the differences
- n : Number of paired observations

Degrees of freedom (df) are calculated as $n - 1$.

2) Accuracy Improvements: **Statistical Significance**

- Mean Improvement: 20.13%
- t -statistic: 12.7680
- p -value: 0.0010
- Cohen's d : 10.5042

Statistical Significance

- t -statistic: 13.0076
- p -value: 0.0010
- Cohen's d : 10.7629

VI. CONCLUSION AND FUTURE WORK

The findings reveal the increased significance of social media conversion metrics on a statistically significant and practically meaningful level when integrated. As is clear from the tables, the p -values for all four conditions were 0.0010. The Cohen values d are greater than 10 for the four studies, indicating high levels of precision and error reduction. Random forest and Gradient Boosting models are specifically useful according to the findings which support the idea of the proposed computational framework that caters to multi-modal data integration.

The results highlight the statistically significant and practically significant impact of integrating social media metrics. With p -values of 0.0010 and Cohen's d values exceeding 10, the enhancements in accuracy and error reduction are evident. Random Forest and Gradient Boosting models are particularly effective, suggesting their suitability for applications involving multi-modal data integration.

Future research should focus on refining feature engineering and model architectures. Advanced methods which will include feature extraction to detect features in audio and social media that were previously undetectable. It is possible to uncover temporal patterns in people's interactions on social media and particularly how engagement varies. Use the cross-headed or composite or integrated models for combining the merits of the different machine learning techniques. These enhancements are crucial to achieve a desirable output.

1) *Data Enrichment*: The dataset can be expanded to improve generalizability. Streaming platform metrics (play counts, skips, etc.) to be incorporated. Extend the intentions of the engagement data across multiple platforms to build up the total level of engagement. Proposition models per genre to mitigate the differences that influence the degree of popularity between genres. Developing proposition models would further enhance the model's predictive accuracy. More data means that the machine can learn far more properly in most cases, which is why data enrichment is crucial.

2) *Technical Challenges and Mitigation Strategies*: Bias in social media data can be managed by removing demographic and temporal bias using normalizations and stratified sampling methods. Dynamic social media trends develop models that will be sensitive to the dynamic nature of level of engagement. Finally, Adaptability which reduce the computational time in case of large amount of data and in order to make predictions at the earliest instance.

A. Broader Impacts

The integration of social media metrics into predictive modeling offers practical implications for the music industry. Created other tools in sync with record labels and artists in order to make the advertising process as effective as possible. Originality: better recommendations to stream services according to the audience. Enhanced perception of how the digital consumption trend is likely to transform the current music trends. Better insights also enable stakeholders to make more educated and smart decisions.

B. Limitations and Study Implications

Several limitations of this study should be addressed in future work. Temporal Constraints, which mean Social media data only assess popularity in the short run, which may not give a background on events in the long run. Demographic variability is also a hassle, as there are variations on the use of engagement patterns across age, regions, and the platforms, which reduce the generality of the model. Genre-Specific Variations means that these popularity indicators may differ greatly depending on genre; therefore, generalized models may not be effective. It is crucial to analyze the limitations that the study brings and address them properly.

However, these are some of the limitations of this study; nevertheless, there is a clear methodological approach that can be used to incorporate metrics derived from social media into song popularity prediction. The results of the study propose the development of a new area of research centered on the interaction of artificial intelligence technologies, music analysis, and consumption of digital media, which expands our understanding of how music consumption processes are developing.

SUPPLEMENTARY CODES

All the codes used in this paper can be accessed through the following link: <https://github.com/L1M1N4L/LearnFODS/tree/main/finalproject>.

REFERENCES

- [1] F. Khan, I. Tarimer, H. Alwageed, B. Karadağ, M. Fayaz, A. Abdusalomov, and Y.-I. Cho, "Effect of feature selection on the accuracy of music popularity classification using machine learning algorithms," *Electronics*, vol. 11, p. 3518, 10 2022.
- [2] H. S. Saragih, "Predicting song popularity based on spotify's audio features: insights from the indonesian streaming users," *Journal of Management Analytics*, vol. 10, no. 4, pp. 693–709, 2023.
- [3] Y. K. Yee and M. Raheem, "Predicting music popularity using spotify and youtube features," *Indian Journal Of Science And Technology*, vol. 15, no. 36, pp. 1786–1799, 2022.
- [4] C. V. S. Araujo, M. A. P. d. Cristo, and R. Giusti, "Predicting music popularity using music charts," *2019 IEEE International Conference on Machine Learning and Applications (ICMLA)*, pp. 859–864, 2019.
- [5] A. Szymkowiak, B. M. Kubala, and M. A. Antoniuk, "Music sales and artists popularity on social media," *International Journal of Marketing, Communication and New Media*, vol. 8, no. 14, 2020.
- [6] J. Lee and J.-S. Lee, "Music popularity: Metrics, characteristics, and audio-based prediction," *Journal of LaTeX Class Files*, vol. 14, no. 8, 2015. [Online]. Available: <https://arxiv.org/pdf/1812.00551>
- [7] R. Díaz-Uriarte and S. Alvarez de Andrés, "Gene selection and classification of microarray data using random forest," *BMC bioinformatics*, vol. 7, pp. 1–13, 2006.
- [8] S. Pahno, J. J. Yang, and S. S. Kim, "Use of machine learning algorithms to predict subgrade resilient modulus," *Infrastructures*, vol. 6, no. 6, p. 78, 2021.
- [9] M. Robnik-Šikonja, "Improving random forests," in *European conference on machine learning*. Springer, 2004, pp. 359–370.
- [10] J. Ali, R. Khan, N. Ahmad, and I. Maqsood, "Random forests and decision trees," *International Journal of Computer Science Issues(IJCSI)*, vol. 9, 09 2012.
- [11] T. Evgeniou and M. Pontil, "Support vector machines: Theory and applications," vol. 2049, 09 2001, pp. 249–257.
- [12] L. Scabini and O. Bruno, "Structure and performance of fully connected neural networks: Emerging complex network properties," 07 2021.
- [13] A. Vasileiadis, E. Alexandrou, L. Paschalidou, M. Chrysanthou, and M. Hadjichristoforu, "Artificial neural network and its applications," 04 2024.
- [14] D. Chicco, M. Warrens, and G. Jurman, "The coefficient of determination r-squared is more informative than smape, mae, mape, mse and rmse in regression analysis evaluation," *PeerJ Computer Science*, vol. 7, p. e623, 07 2021.